



Nacreous vs Tropospheric

An array of photos showcasing the difference between nacreous and tropospheric clouds.
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C Nacreous Clouds

While they say every dark cloud has a silver lining, the beautiful “mother of pearl” nacreous clouds have a dark side.

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ost of the clouds that we see in the sky are within an altitude of 10 kilometres, which is the height of the

troposphere, the lowest and densest layer of the atmosphere. Above it, is the stratosphere which extends up to an altitude of 30 kilometres. This atmospheric layer is much drier than the troposphere, and there are relatively few clouds. When the temperature in the stratosphere reaches below the frost point (-78°C), whatever water vapour is suspended in the air condenses into particles of

ice. Such temperatures for the most part, occur only in winter. When this happens with the wispy stratospheric clouds, they form what are known as polar stratospheric clouds (PSCs), or nacreous clouds. These clouds are very rare, and can be seen in the northern latitudes, including Canada, Scotland, Northern Russia and the Scandinavian countries. They are, however, more common near the South Pole, where the winter temperatures in the stratosphere are more likely to fall below the frost point. The temperature of the stratosphere increases with the altitude, which means that the nacreous clouds are formed only in the lower altitudes.

They appear as rolling walls made up of rainbows, because the ice particles scatter the light all across the spectrum. The optical phenomenon is similar to the colours that you see reflected off a DVD. The ice particles themselves are very fine, which causes the light from the Sun to be scattered, or refracted into so many colours. Because of their high altitude, the clouds are actually lit up from below by the Sun.

The colours become richer as the Sun goes further beyond the horizon, as the altitude allows the clouds to reflect the sunlight. Even when the rest of the sky is a dull, dark grey, these clouds can flare up in rich shades of orange and pink. When appearing at the same time as lower altitude tropospheric clouds, their great heights are visible, and are a stark contrast to the grey, featureless clouds closer to the surface. The phenomenon, known as cloud iridescence is seen when the Sun is between one and six degrees below the horizon, or the period of civil twilight. During the dawn, the process is reversed, and the clouds become less colourful as the Sun rises. They are called mother of pearl clouds because of their characteristic sheen.